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Week-1

Dr. Ashwin J. Baliga.
Dr. Aditya Chandel.
Dr. Siddarth Arora.
BDM Capstone Portal. ? Project.

Price Elastic & Price Inelastic

Account & finance are language of data. Meaning to data
↓ ↓ ↓
How Why what is happening

Books { Tata Macro Hill
Bank of economics & Microeconomics.

quantitative finance (adv)
quantitative trading firms.
options & futures.

Corporate finance.

BDM (why, what needs, How) ?
↳ Bussiness
↳ Data
↳ Management.

Data Centric Solution.

Elastic: Those who can do tradeoff./ change.

Inelastic: Those who can't

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Why to study Economics?

Simplify the word Scarcity → Human wants → unlimited.
→ limited Resources.

Need to Allocated Resources.

To fulfill Human wants.

Optimisation needs.
↳ objective fxn → Get 100% accuracy
↳ constraints. → finite computation, finite memory
↳ finite time, finite computer.

Economics = allocation of resources which we are going to optimise with limited resources.

Economy & Economics.

	Economics	Economy.
Nature.	Academic field / Social science	Real world system. Actual activities & interactions.
Focus	Theories, models, analysis.	What is happening.
Question it tries.	Why & how do these things happen?	
Example	Studying why inflation occurs. Involves looking at the society from bottom up.	India's inflation rate this year.
		up to bottom. / top down.

Two branches of economics

Microeconomics: invidual decisions, such as how consumers choose, how firm set prices, and how markets for specific goods work.

Macroeconomics: the whole economy, covering GDP inflation, unemployment, interest rates, and govt. policy.

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Economics basics

Needs, wants and desire

Needs are essentials for survival and well beings.

Wants: Wants are known to enhance the quality of life. e.g. entertainment, travel etc.

The diff. between wants and desire is a form of affordability.

infinite wants / insatiable wants.

How to produce goods & services?

Resources → tend to be scarce in nature (short in supply (shortage).

Human wants are unlimited (infinite) but the resources to meet these wants are scarce.

» The scarcity of resources implies that they are constraints (limitation) on goods and services that can be produced, and what human wants could be ultimately satisfied.

Economics is the science of constrained choices.
Economics is the science that deals with allocation of scarce resources to meet the unlimited human wants.

Economists classify the good into 2 types.
① Economic goods
② free goods

Economic goods - are those goods that are scare in nature.

Free good refer to those goods where there is enough supply to meet the human want. Eg: air

★ If people do not pay for goods, are they free goods?
NOT NECESSARILY

Free hospitalisation: This is ecomic good and not a free good, because you allocate the scarce resources to provide healthcare.

To produce goods and services, we need what we call as "resources".
↳ we get from mother earth.

Economists divide earths scarce resources into four factors of production:
- Labour
- Land
- Capital
- Enterprise

Production any activity carriants that satisfy a human want.

Land refers to scarce resources such as oil, sea etc.
Labour: who are available to work.
Capital: good that are used to produce other goods. E.g. buildings and machinery

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Enterprise: say entrepreneurs, who take 3 factors → land + labour + capital to produce goods and services.

All these when combined, results in production of goods and services that meet the human wants.

When these goods and services are supplied, we say production has taken place.

Economic problem:
Human wants are infinite, but the resources are "scare"

So what happens due to this

We cannot get whatever we want and thus there is a need to carefully do an allocation of scarce resources.
We refer to this as "economic problem" and economics is all about tackling this issue. .....

Ost common misconception is that economy is money. While economists study money as a topic. it is done so as it addresses the economic problem.

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Also what happens due to this

We cannot get whatever we want and thus there is a need to carefully do an allocation of scarce resources.
We refer to this as "economic problem" and economics is all about tackling this issue. .....

- Most common misconception is that economy is money. While economists study money as a topic. it is done so as it addresses the economic problem.

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① Business success is guranteed if you know economics another common misconception.

Branches:
Microeconomics. → smaller parts
Macroeconomics. large economy/entire.

Demand: desire backed by the ability and willingness to pay

A person from a low income group wants to buy a luxury car. This cannot be considered as a demand by the company, because the person has no purchasing power.

Anyone without purchasing power cannot demand a good/service.

Economics is rooted in fundamental concepts such as scarcity, utility and marginality.

Individuals/households sell their labour to firms and earn money.

Firms in turn are dependent on individuals who in turn produce the items, the firm sells.

Scarcity: It means that there is short supply. Unlimited wants and scarce resources kind of creates an imbalance → scarcity.

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Theoretically speaking the human wants are unlimited / unlimited human demands.

Resources are scarce - Land, capital, Labour, production, technology, knowledge, time etc..

What are the causes of Scarcity?

1. Demand - induced Scarcity
2. Supply - induced Scarcity
3. Structural scarcity

Demand induced Scarcity: The supply remains the same, however the demand for the resources are increased / goes up. When the population goes up or demand for a particular product is increased.

Supply induced Scarcity: The demand remains the same, however, the supply reduces.
e.g. Recent geopolitical tension; Crop failure
force majeur.

Structural Scarcity: When some people tend to have less access to resources in comparison to others.

e.g. People living in Rural areas / villages tend to have lesser amenities (healthcare, passport etc).

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Utility

In economic utility refers to the satisfaction consumer gets upon consumption of goods / services.

This utility influences the economic decisions for instance, 2 people buy a coat. It is very unlikely they will experience the same utility.

PersonA: may use this more frequently.
PersonB: may use this less frequently.

Marginality:

People don't make the economic decision all in one go, but they do make it bit by bit.

Law of diminishing marginal utility.

Marginal Utility:

The incremented increase in utility that is derived from the consumption of one additional / extra unit.

As the consumption increases, the marginal utility associated decreases come down.

Law of diminishing margin utility makes several assumptions

① The items that are consumed are identical
② units / items are consumed quickly and there are a few breaks in between.

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④ The consumer can make rational decision with respect to consumption.

④ The items can be measured.
⑤ The consumers taste remain constant.
⑥ The units that are consumed is neither too big / nor too small.

Rationality.
Human beings are rational animals.
Cost benefit analysis / assessment

They tend to make choices when benefits > costs (benefits outweighs costs)
And may not make decision when costs > benefits.

Opportunity Costs.
Dan Ariely] Every time when you spend money on something, you are not able to spend it on something else.

When the consumer make a choice, they have to / need to give up / reject the other choices. The costs associated with such lost opportunities is known as "opportunity cost".

» Produce vs Outsource.
Specialization.
In most countries, people tend to earn money by specializing in particular tasks such as farming nursing etc. This is also known as division labour.

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Economic Sectors.

Primary
Secondary
Tertiary
Quaternary.

Primary → workers in primary sector extract raw materials from the earth. Fishing, farming, mining etc.

Secondary → involves conversion of Raw material into finished product. Vital utilities like electricity and fuel.

Tertiary → Servicing sector.

Quaternary → Scientific Research & development, R&D. Education, public.

Economy System / Economy:
↳ Classification
↳ Command → Societist
↳ Market → Cap.
↳ Mixed → India

① Command / Societist / Control directed Economy:
Rule based System
Central authorities make rule what to produce, when & How to produce.

If demand > 100 kg/Day:
Produce (150)
else:
halt production

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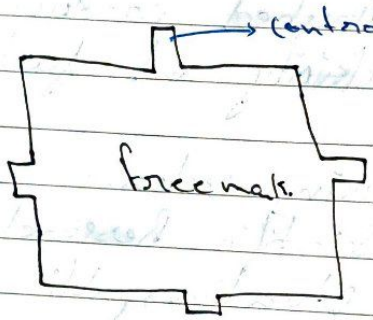
Free Market Economy. (Capitalism)

Mixed Economy.

Companies authorities make decision but somewhere govt also interfares so its Mixed economy.

10 min delivery → by authorities but ~~~~
accident & work life balance →

NPCI - UPI transactions.



Command: 90's → license Raj.

Free : TCS, Zomato, Trading Crypto

Mixed : Mordern India.

Classification
↳ open / close economy.

Open : Anyone can come Invest / trade.

Close: No exchange out of the boundry.

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Economy as Social Science

because it studies human behaviour specifically how invidiuals, firms and governments make choices when resources are limited.

Limits..

- Human behaviour is less predictable than physical laws.
- Values and ethics influence which questions get asked (positive vs normative economics)
- Results often depend on context, culture and time period

Product Chain Analysis..

90's ↙ Timely we need it. (ancestors) ↓ Item sticks ↘ Demand ↗ wants. ↳ Technological development

Toasting of trees

Late 90's → Normal toothbrush

20s → Zigzag.

21st → Electric toothbrush.

OralB → Production (Limited raw material) → Delivery ↳ Distribution ↳ Vendors/Supplier ↳ Shop. ↳ Customers.

Optimisation
↳ Objective fxn.
↳ Constraints

ex. Ramu Tea stall.

Tea → 6₹ → 4Rs/cup.
filter coffee 12₹ → 6Rs/cup.

Total Budget.

$6T + 12f = 1200$ —①

Order 40 FC
↳ Take & execute.
↳ Say No.

$6C = 1200 - 12 \times 40$

$6 = \frac{720}{6}$

$= 120$ cup of tea.

Graded Assignment

④ $100\xi_2 + 400\xi_3 = 20,000$

$100\xi_2 = 20,000 - 4000$

$\xi_2 = \frac{16000}{100} = 160$

⑧ $160 \times 500 \quad 10 \times 1500$

$80,000 + 15,000$

Total.

Case 2: Basmati Rice — {10,000 litre/day} — ₹ 15000/acre
Wheat — ₹ 8000/acre.

BR: 500 litre/day/acre
W: 20 litre/day/acre.

① $500B + 200W = 10,000$
② maintain the fertility/nitrogen content in soil
↳ 5 acre cultivate Basmati Rice

$500 \times 5 + 200W = 10,000$
$W = \frac{7500}{200} = 37.5$ acre.

Profit $75000 + 37.5$
$75000 + 3,00000$
3.75 lakh.

Elements of Economics.
① Production
② Distribution
③ Consumption.

What/How to produce. then Whom to produce. } Problem of economics.

Market: exchanging the item/accomodities ⇒ Money flow

* Fulfil needs.
* Benefits.

① Production
Raw Material → Utilise item.

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Outcome of Products

Visible	Invisible
Tangible	Intangible
→ touch, store	→ Services
	→ only experience / can't store.

Factors of Production

→ Land
→ Labour
→ Physical Capital → fixed Capital → Long term investment
→ working Capital. → short term / monthly
→ Entrepreneurship / Human capital.
↓
salary / electricity bill

Land + Labour + Physical Capital
⇓
Manage these resources.

Primary Sector
↳ Raw Material

Collection of Raw material → Primary Sector
↳ ① Natural Resources
↳ ② Raw material.

Red Collar Workforce.
approx. 17% of GDP

Secondary Sector.
Production activity → all process started.

Blue Collar workforce
27% of year GDP.
↳ 14% manufacturing sector.

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Tertiary sector.

Servicing sector after checking.

↳ Logistic / supply chain / quality management

» 54% of year GDP.

Quaternary Sector.

Research & development, Education, Public, intellectual services.

There is no single official percentage for quaternary sector's contribution to India's GDP because it is not seperately classified in standard national accounts.

Week 2:

Market: is a group of buyers & sellers of a particular good and services

buyers as a group determine demand
sellers as a group determine supply.

Organised Markets:
→ highly organised, agricultural commodity.

Unorganised Market:

Forms of Market:

① **Competitive market:**
→ An invidual has a negligible impact on the market price

② In a perfectly competitive market, two characteristics:
→ Goods offered for sale are the same and.
→ B&S are so numerous that no single buyer or seller can influence the price.

② **Non-competitive markets.** E.g. Monopoly market, Oligopoly market.

Monopoly: A single seller set the price. IRCTC, Indian oil corp.

Oligopoly: A market dominated by a small number of large firms. e.g. Telecom Industry

Demand.

Key words are:

Willing: the customer wants the product.

able: the consumer has the purchasing power.

different price: demand changes with price.

given period of time: demand is measured over time.

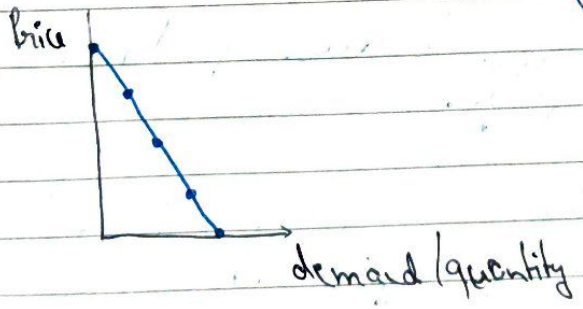
- Demand has to be backed by purchasing power.

The law of Demand: This is the prevasive economy principal stating that, other things being equal, when the price of a good rises, the quantity demanded of the good falls. & vice versa.

A Demand Schedule is a table that displays the relationship between the price of a good and the quantity demanded.

Demand Curve: is a graphical representation of the demand schedule. It slopes downward because the lower price naturally increases the quantity demanded.

$$D = f(O, P)$$



Market Demand Versus Individual Demand:

Market demand is the sum of all individual demands for a good and service.

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Quantity demanded in a market is the sum of the quantities demanded by all the buyers at each price.

- Market demand curve demonstrates, how the total quantity demanded varies as the price of the good changes, while holding all factors constant.

- Hence the market demand curve is found by adding individual demand curves.

Supply

The quantity of a good or service that producers are willing and able to offer for sale at diff. prices during a given period of time.

» willing producer want to sell
» able capacity / resources to produce.
» diff prices supply usually changes with price
» given period of time given period of

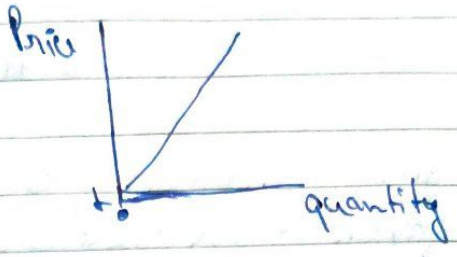
- Higher prices higher supply because producers are motivated by profit.

Law of Supply: price of a good rises, the quantity supplied also rises, vice versa.

This happens because selling becomes more profitable at higher prices, prompting seller to produce more.

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Supply Schedule → tabular form.
Supply Curve → relationship b/w price & quantity



Market Supply:

Shifts in the Demand Curve:

- The demand curve shifts when a variable that affects buyers' desire to purchase alters the quantity demand at any given price.

- An increase in demand refers to any change that raises the quantity demanded at a price point, shifting the demand curve to the right.

  A decrease in demand reduces the quantity demanded at every price, shifting the curve to the left.

Key variable that shift the demand curve include:

- Income ↑ demands for normal goods ↑
  If demand rises when income falls, its an inferior good.

- Prices of a Related Goods: Two goods substitute if a fall in the price of one reduces the demand for the other.

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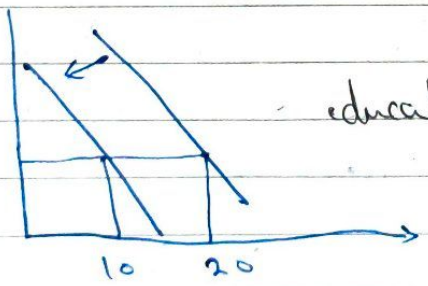
- Two goods are complement then.
  » If fall in the price of one increases the demand of other.
  (Hot fudge & ice cream, often consumed together)

- Preference / Tastes: Positive preferences increases demand.
  • Fitness trend increases demand for protein bakes

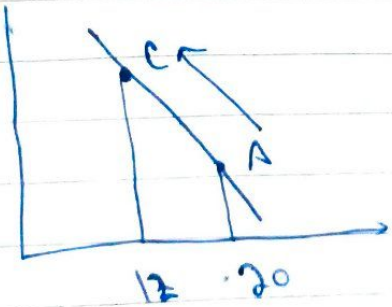
- Future Expectation: Expection of fuel price hike increases petrol purchase now.

- Number of Buyers: An increase in the number of consumers will increase market demand.
  e.g. Rise in demand for real estate if a company is setting up a 2000 person facility in a location.

Case Study: Ways to Reduce Smoking Demand.



educating people.
no price hike



tax raises / Price hike.

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Supply Curve

- Any factor that changes the quantity supplied at every price will cause the supply curve to shift.
  An increase in supply shifts the curve to the right and a decrease in supply shifts it to the left.

- Key variable that shift the supply curve include:
  - Input prices: If input price rise, production becomes less profitable, and firms supply less.
  - Technology reduce labor costs & increase the supply.
  - Expectation: If it expected the market price rise in future then company may store the current production, reducing the immediate supply available to the market.

Number of Sellers: If sellers leave the business, the market supply will decrease.

Supply and Demand Together: Equilibrium

- Equilibrium is the state where the price has reached the level where quantity supplied equals the quantity demanded. Supply and demand curve intersect.
- Equilibrium price that balances the quantity supplied and demanded.
- Equilibrium quantity supplied and demanded at the equilibrium price.

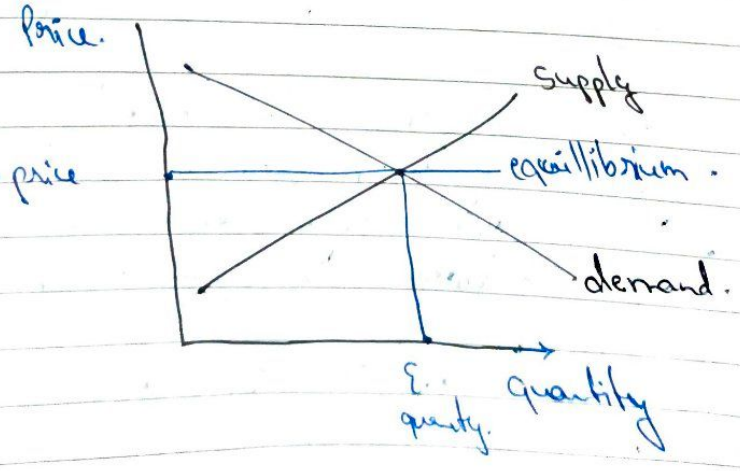
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Surplus: If the market price is above the equilibrium price, a surplus (or excess supply) occurs because the quantity supplied exceeds the quantity demanded. Sellers respond by cutting price to increase sales, which pushes the price back toward equilibrium.

Shortage: If market price is below the equilibrium a shortage (or excess demand) occurs. because buyers want more than is being supplied. Sellers can respond by raising price without losing sales, which drives the price up towards the equilibrium.

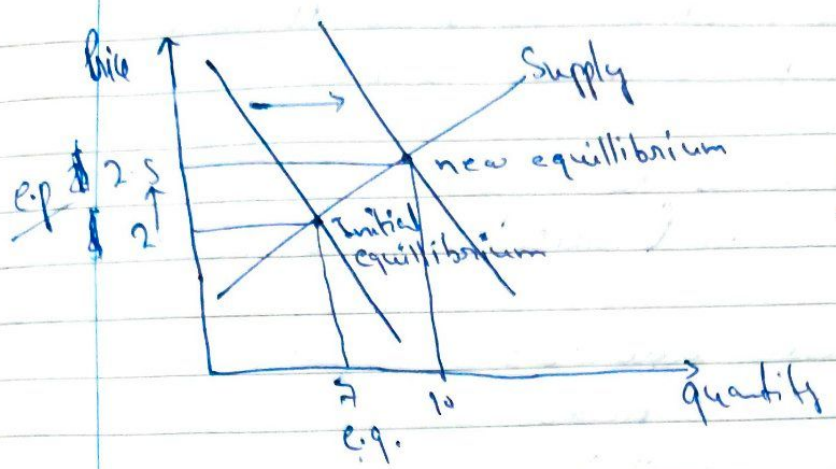
Law of Supply and demand: Claims that the price of any good naturally adjusts to bring the quantity supplied and quantity demanded into balance.

Equilibrium - Graphical Explanation



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How an increase in demand affects the equilibrium.



price is reason

Demand Supply

- Elasticity measures the degree of responsiveness of one variable to change in another variable.
- Price elasticity of demand measures how much the quantity ~~responds~~ demanded changes as a response to a change in price.
- Demand is considered elastic or highly elastic if the quantity demanded responds substantially to a price change, and inelastic or less elastic if it responds slightly
- Example:
  - If the netflix raises price and many customers cancel subscriptions: Demand is elastic.
  - If petrol price rises but people still buy it demand is inelastic.

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Determinants of Elasticity

There are no universal rules for determining elasticity (but general rules include:

- Availability of close substitutes:
  eg. (butter & margarine)
- Necessities versus luxuries:
  inelastic — elastic
- Definition of the Market: Narrowly defined market have more elastic demand than broadly-defined markets (like food) because it is easier to find close substitute for narrowly defined goods.

Time Horizon: Goods tend to have more elastic demand over long time horizons.
eg. Change in Petrol price, demand falls only slightly but over years people buy more fuel efficient cars.

Computing the Price Elasticity of demand.

Price Elasticity in the Demand = $\frac{\text{\% change in quantity demanded}}{\text{\% change in price.}}$

$\frac{\Delta Q}{\Delta Q} \times 100$

$E_D = \frac{\Delta Q}{\Delta P}$

- Because Quantity demanded is negatively related to price, price elasticities of demand are naturally negative numbers. However, it is common practice to drop the minus sign and report all price elasticities of demand as positive numbers. (absolute value).

Slope = $\left(\frac{1}{Q}\right)$ × $\left(\frac{\Delta Q}{\Delta P}\right)$ → slope of demand curve

variable

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Unit Elastic demand parameter.

% change in demand equals % change in price.

Perfectly Elastic demand:

Even a tiny increase in price causes % change in price.

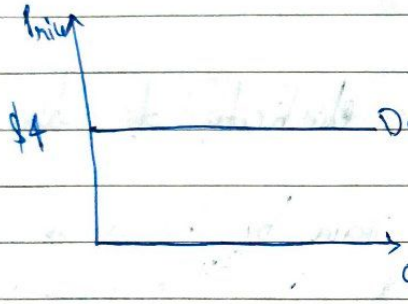
Perfectly Inelastic demand

Demand remains constant regardless of price changes.

Elasticity Curves

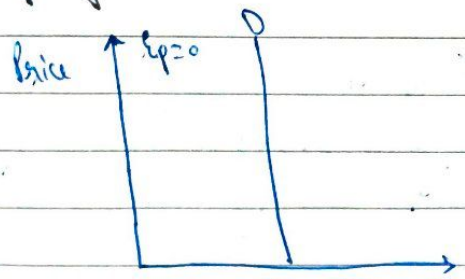
Perfectly Elastic Demand

elasticity = ∞



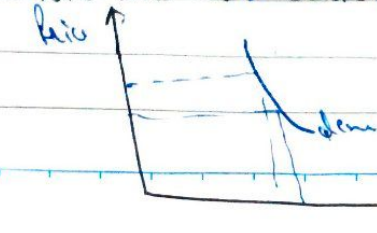
Perfectly Inelastic Demand.

e = 0



Inelastic Demand: elasticity is less than 1.

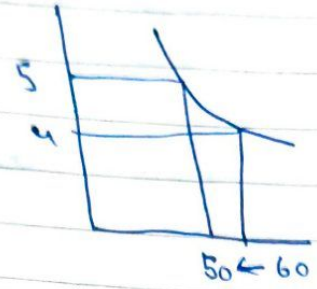
Price ↓↓ = TR ↑↑



$0 < e_p < 1$

- less sensitive
- die heart fans.
- Rise = TR

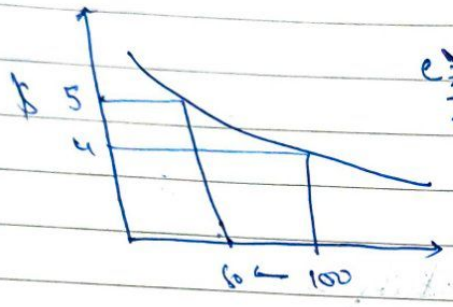
Unit Elastic demand.



- Change in price = no effect on revenue.

$e = 1$

Elastic demand



- consumers are sensitive to price change

$e > 1$

- lower price TR ↑

Total Revenue and the price elasticity of demand.

- Total Revenue is the amount paid by buyers and recieved by sellers, calculated as the price of the good (P) times the quantity sold (Q) = PQ.

- If the demand is elastic. The price and total revenue move in opposite directions because the fall in quantity is proportionately greater than the rise in price.

- If demand is inelastic. The price & total revenue move in same direction because the fall in quantity is proportionately smaller than the rise in price.

- If the demand is unit Elastic, total revenue remains constant when the price changes.

The Elasticity of Supply.

Price Elasticity of Supply measures how much the quantity supplied changes in response to the change in the price.

- Supply is elastic if the quantity supplied responds substantially to price changes, and inelastic if it responds only slightly.
- This depends heavily on the flexibility of sellers to change the quantity of supply.
- Supply becomes more elastic when:
  - firms responds quickly
  - resources are available
  - production adjustments are easy.
- Supply inelastic when:
  - Capacity is fixed, or
  - production takes time
  - resources are difficult to obtain.

Key Variable that impact Supply.

- Longer time increase elasticity because firms can adjust production.
- Availability of spare capacity: More unused capacity increases elasticity.

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Key Variable that impact Supply.

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- Availability of spare capacity: More unused capacity increases elasticity.

Availability of Inputs/Resources:
Easy access to labour/raw materials increases elasticity.

Nature of the Product:
Perishable or unique goods tend to have inelastic supply.

Technological Capability:
Better technology improves ability to scale production.

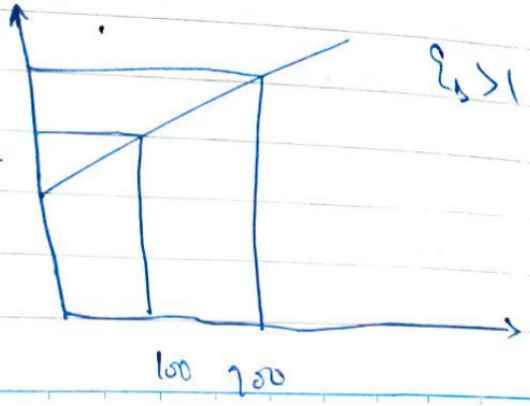
Govt Regulation:
Heavy Regulations reduces elasticity.

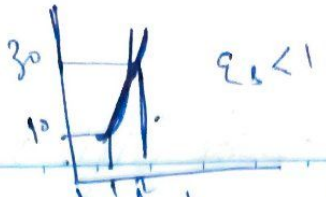
Computing the Price Elasticity of Supply.

Price elasticity of Supply = $\frac{\%\text{ change in quantity supplied}}{\%\text{ change in price.}}$

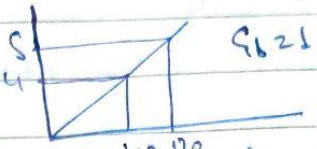
Supply Elasticity Classification

Elastic Supply:
A small change in price leads to a large change in quantity supplied



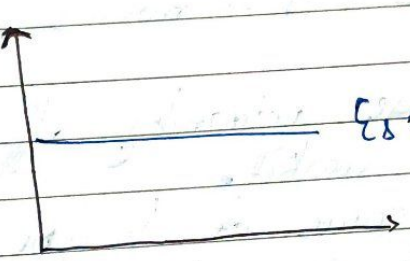


Inelastic Supply:
A change in price leads to a relatively small change in quantity supplied.

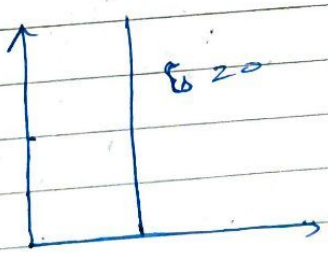


Unit Elastic Supply
Percentage change in supply equals % change in price.

Perfectly Elastic Supply
Suppliers are willing to supply any quantities at one fixed price, but none at lower price.



Perfectly Inelastic Supply
Supply remains constant regardless of price changes.



Can Good news for farming be bad news for farmer?!
Sometimes

Case Study: Why did OPEC fail to keep the price of oil high.
relatively inelastic.

Problem A shift in the supply curve leads to massive increase in price.

- long run: Producers outside OPEC responded by increasing oil exploration and building new extraction capacity.
  while consumers respond by conserving energy.

- Because long run supply & demand curves are much elastic, the same reduction in supply by OPEC eventually caused a much smaller increase in the price of oil, causing prices to steadily decline in the 1980s.

Economies of scale & economies of scope.

- Economies of scale primarily affect the supply side of the market.
- When firms produce at larger scales:
- their average cost of production falls (as the fixed costs are spread over larger units),
  - profitability increases, and
  - firms become willing to supply more goods at lower prices (competitive advantage).

- This causes the supply curve to shift to the right.
- Economies of scope also affect the supply side, through product diversification & resource sharing.

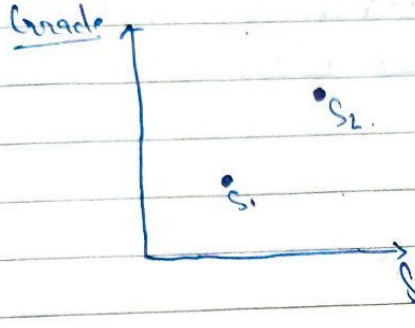
When firms produces multiple related products together:
- they reduce production costs
- improve operational efficiency
- increase supply capability across product categories.

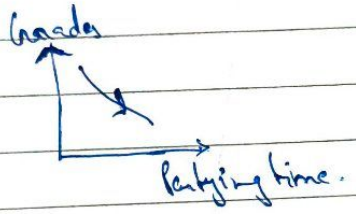
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Live Session 1:

Graphing: When you are plot anything variables are dependent.

Dynamic Pricing: Changing ...





Profit = Total revenue − Cost of production.

⇓

Quantity × selling price.

Production function: Relationship btw your input & output.

case: Farm. — max outcome 2.5 Tonnes of wheat.

$Pf = f(K, L)$ → Land, Labour

$q = K \times L$

max amt of wheat

$q = X_1^{\alpha} X_2^{\beta}$ # Cobb-Douglas equation

→ Total production/output.

$Y = A L^{\alpha} K^{\beta}$ Cobb Douglas equation

→ Total factor productivity

L : labour input
K = capital input
α : } Constraints. output elasticity of labour
β : } output elasticity of capital.

Short Run production:

input : Land & labour
(fixed) (variable)

$q \neq f(L)$
$q = f(\text{labour})$

$q = f(K, L)$ → fixed

$q = f(K)$

Long Run production.

Every thing is variable.

for everyday. Change in output you are getting.

No. of worker	Output (quantity/hr)	Marginal Product (MP)	Cost of factory (fixed)	Cost of labour	Total Cost
0	0	-	₹3000		3000
1	50	50	3000	₹800	3800
2	90	40	3000	₹1000	₹4000
.	.	.	.	.	
.	.	.	.	.	
.	.	.	.	.	
6	155	5		₹3000	₹6000

Marginal Production:

MP of any input in production process
↳ change in the quantity of output.

$$MP = \frac{\Delta Q}{\Delta I \rightarrow \text{worker}} = \frac{90-50}{2-1} = 40$$

* Law of diminishing Marginal Return / Production / Diminisal Marginal Productivity.
As you add more of a variable input (like labour) to a fixed input (like machinery or a building) the additional output generated by each new unit of input will eventually decrease.

Cost → Total Cost:
Avg Cost: (ATC) = $\frac{(TC)}{(Q)}$
↳ Marginal Cost. = Limit Variance par extra Kitna aaya.

(1)

* TC = Total Fixed Cost + Total Variable Cost.

Fixed Cost: Factory Rent: Insurance, Salary (if number of employee ↑ then its variable also dependent)

No of workers ↑ fixed cost ↓

Constraints:

AFC

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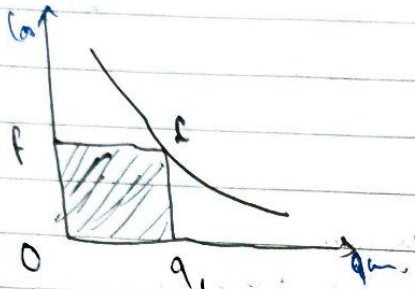
Never Zero

② Variable Cost :

③ Avg Cost : Per unit cost of output.

$AFC = \frac{TFC}{Q} \Rightarrow$

$AC = AFC + AVC$



$TFC = AFC \times Q_m$ = Area of rectangle

$= OF \times OQ_1$

Output (q)	TFC	TVC	TC	AFC	AVC	MC $\frac{\Delta T}{\Delta Q}$
0	20	0	20	—	—	—
1	20	10	30	20	10	10
2	20	18	38	10	9	8

given.

Opportunity Cost: → also called ghost path. by opting other path

» You chose one but lose another.

Sunk cost: Whatever is not getting recovered is called sunk cost.

Limited Resources — Choices ⇒ Tradeoff.

Constraint = Data Science

» Time
» Resource
» Data
» Privacy

Scarcity + Prioritise Satisfaction ⇒ Constant Tradeoff.

Case: ⇒ Fraud detection in Bank.
Target: AI Model ⇒ Predict
↳ Applied for loan ⇒ default or cap. for loan

Constraint = Limited time of 1 month to build model
Dilemma = multiple ways.

option 1: Complex deep learning model
option 2: Simple logarithmic Regression model.

	Option 1	Option 2
accuracy	95%	85%
time req.	4 weeks	1 week.
comp. costs	Expensive bPU Lag storage High computation cost	Minimal.
	Black Box.	Highly interpretable

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Profit Illusion:

@ Economic Profit ⓑ Accounting Profit.

Economics (Opportunity cost)
- Explicit (Actual money flow, Investment in day to day living.
- Implicit (Actual time associated (Consume time)

→ Invisible / intangible.

Explicit cost	Implicit cost (value given up).
Wages / office Rent Raw material Hard / computing cost. Software license Replacement cost.	o Value of time and efforts like you are to open / build your own company.

Economist	Accountant
* How firm / Business make production & pricing decisions.	Tracing money flow in and out of the business.
* Cost Analysis.	Explicit → financial Transaction Pay bills. ↳ out of Pocket money
* Include the value of next alternative which you are sacrificing	—
* Based on your actual own & loan and Capital Cost	Based on tax rules and accounting standards.

Total Revenue	Economist	Accountant
	Economic Profit + Implicit Cost + Explicit Cost	Accounting Profit + + Explicit Cost.

Profit = TR - Cost of production. (i)

P = TR - Cost (ii)

TR = Profit + Cost. (iii)

★ Economic Profit : TR - (Explicit cost + Implicit Cost)

Accounting Profit : TR - (Explicit Cost)

Case: X - Salary 15 lpa | founded 00s data → 25 lakhs in 1 year.

Exp. => Rented Co-Worker space = 1.5 lakh
Software licensing = 1 lakh
Cloud computing = 2 lakh.

Total Explicit Cost = 4.5 lakh.

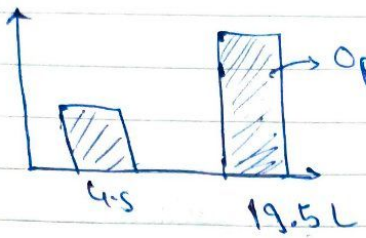
Accounting Profit = 25 - 4.5
= 20.5 lakh.

Economic Profit =

Implicit cost ⇒

Salary = 15 lakh.

Economic Profit = Total Revenue - (Explicit + Implicit cost)



EP = 25 - (19.5)

EP = 5.5 lakh.

AP = 20.5 lac.

Summary ⇒

TR = 50K pm Explicit Cost = 20K pm

Implicit = 50K pm

Accounting Profit = 30,000 | EP = 50000 - (20000 + 50000) = -20,000

-ve sign means You are not going to sustain for a long time in Market.

Three possibility

1. EP = 0 Stable / Equilibrium
2. EP = +ve opportunity to expand your business
3. EP = -ve exit from market.

Case 2: Anya.

Rs: 1500 → Explicit Cost
Rs. 4000 → Total Revenue.

₹2000 - Implicit Cost.

EP	Accounting Profit
EP = 500	AP = 2500

Production Possibility Frontier / Transformation possibility curve
Production Possibility Curve.

e.g.

Land	→ Power Bi
Machin	Cycle, Toys.
Worker	→ AI toolbox fraud detection

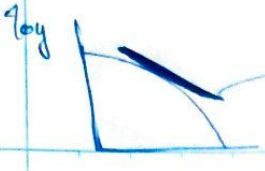
PPF is a graph representation ⇒ ?
Possible combination of goods / services / products.

⇓

To achieve maximum economic efficiency / max utilization
Economic growth. / of available resources.

Trade-offs. Toys ↓↓↓↓ —— Cycles ↑↑↑.
(2s) (2s)

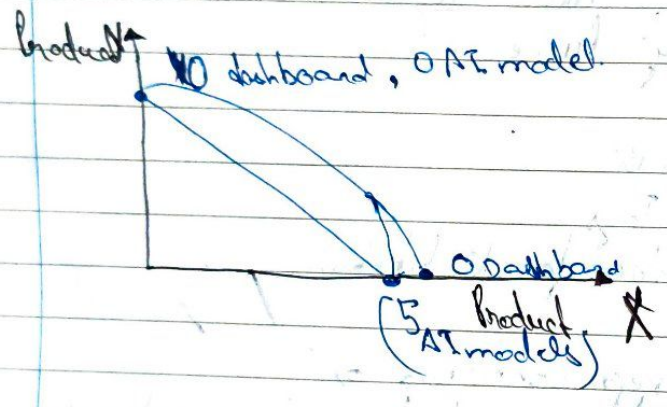
Input → Output (max Resource utilisation)
Land/Labour/machine → as per demanded
making diff. combination of Goods/services



Slope = Opportunity Cost.

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Product X => Custom AI models	Product Y = Dashboard.
Notes: supply chain optimisation, fraud detection. Dynamic pricing models.	To showcase the live data and analysis.
Resources: High	Low
Unit Price: 15.0 lac.	₹2.5 lac.



Points	X	Y
A	0	40
B	1	38
C	2	32
D	3	25
E	4	14
F	5	0

$B = 1x + 38y \Rightarrow$ Max efficiency

$E = 4x + 14y =$

Marginal Rate of Transformation. (MRT).

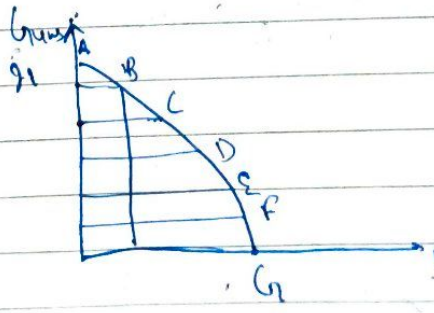
$$\frac{\Delta y}{\Delta x}$$

Assumption:
- Input fixed
- Resources fully utilised.
- Tech + other conditions → fixed.

Marginal Opportunity Cost.
What are you sacrificing for the what you gain.

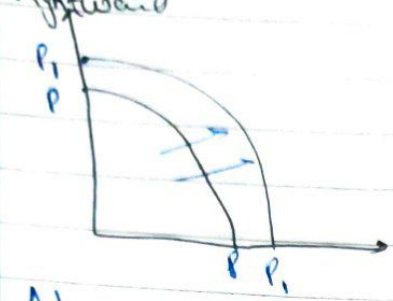
$$MOC = \frac{\Delta S}{\Delta G} \rightarrow \frac{\text{change in Sacrifice}}{\text{change in Gain}}$$

Possibilities	Guns	Cheese	MOC	MRT
A	21	0	-	~~1S : 1G~~ -
B	20	1	1	1S : 1G
C	18	2	2	2G : 1S
D	15	3	3	3G : 1C
E	11	4	4	5G : 1C
F	6	5	5	6G : 1C.
G	0	6	6	



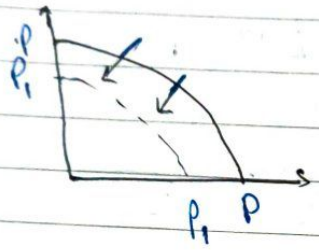
Shift in PPF

Rightward



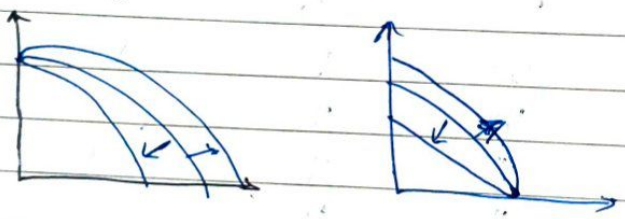
↳ Advancement & Tech upgrade
↳ Growth of resources

Leftward Shift



- Scarcity in manpower
- Production falls.

Rotation in PPF



Change it as for requirement.

PPF ⇒ Concept of efficiency = $\frac{\text{output}}{\text{input}}$

efficiency = 100% No wastage ⇒ full utilization of Resources

> 100 over Utilization of Resources

< 100 under utilization of Resources.

- Possibility of Trade off
- Achieving efficiency maximum
- Evaluate opportunity cost.
- Planning Economic Growth

(4) ATC = 1500 per chip
MC at 10,001st chip is 0.50.

ATC will decrease

$TC = ATC \times Q$
$TC = 1500 \times 10,000 = 15,000,000$

$MC_{new} = \frac{TC + MC}{Q + 1}$

$= \frac{15\,000\,000 + 0.50}{10001} \simeq 1,499.93$

Remember :
MC < ATC ATC decrease
MC > ATC ATC increase
MC = ATC ATC remain unchanged.

(6) $AFC = \frac{TFC}{Q} = \frac{TFC}{Q_2} = \frac{2(TFC)}{Q}$ (AFC)

(7) ~~ATC = AFC + AVC~~
~~15,000 = 5000 +~~

$4 \times 2000 = \frac{35}{2000}$
4×2

$TVC = TC - FC$
$= 70000$

$AVC = \frac{TVC}{Q} = 35$ /unit.

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(2) By choosing option A :

Total profit = 500000 earned − 1.5 lacs penalty

= 3.5 lacs.

option B

Saving → 10K per month

= 60, K.

Also avoids 1.5 lac Vendor extension penalty

Note: The 1.5 lac. penalty is an additional cost of choosing option A, not the opportunity cost itself.

$ & ₹ are worth using twice.

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Week-3 Understanding Business Basics.

Customer and a Consumer.
"A customer can be a consumer ; a consumer need not to be a customer."

Marketing vs Sales.

Marketing → AMA (American Marketing Association)
"Choose a target market, get, keep and grow customers, create, communicate and deliver an attractive value propositions."

Marketing → awareness → cashout flow

Marketing = 5Cs + 4Ps + STP

4Ps = product, price, promotion and Place.
5Cs = Customer, Company, Context, Competititon. and Collaboraton.
STP = Segmentation, targetting and positioning

Segmentation : group of customers having similar needs and wants.

Targeting is based on the mode - frequency is higher.
Positioning → how do you position this in the mind of consumer?

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Sales: Cash inflow
Selling.

To sell we needs minimum 2 entities:

Seller / supplier / Service providers → Buyers / Customer / client.

The seller sells the "offering" to the buyer at a "profit".

Offerings: Products / services / product and services.

Cost Price: Cost incurred in making a product:
Selling price: Cost at which product is sold in the market = 25 rupees.

SP > CP ⇒ Profit (1st Preferance)
SP < CP ⇒ Loss (Last Preference).
SP = CP = No P / No loss. (2nd preference

B2B & B2C

Business to Business & Business to Consumer.
To sell we need minimum of 2 entities

Seller / supplier → Buyer / Customer
The buyer or customer is the individual.

exapl. Amazon (Business) → Ashwin (Buyer)

In B2B, the seller & Business both are an business firm / organisation.

Supplies → Amazon → Customer.

Wholeseller 4 retailer.

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Product vs Service → then is no ownership of the physical element involved.

Services are → HITP → Intangible → can't see 4 touch.
→ Heterogeneous
→ Inseperable
→ Perishable

ex. Consultancy, education, banking, logistics, etc.

Why is that ex. need to study services?

1. It contributes a lot to the GDP (80%. GDP)
2. A lot of employment is generated in the services sector.
3. Services provide the company with competitive advantage.

Pure Services 4 add on services

Differences between B2C 4 B2B

B2C	B2B
• End Customer is individual	End Customer - Business.
• Number of Customers - high	Less and they are geographically concentrated.
• Supply chain - Larger	Shorter
• Purchase duration - Shorter	Longer (3 months → 1 year).
• Purchase value - Less	Purchase value - high. (millions to Billions INR.$)
• Transactional relationships	long termon Collaborative.
• Price Elasticity	Price Inelastic. (conditional).
• Advertisement 4 promotions → Higher.	lower
• Agreements 4 contracts - less	Higher
Decision maker → Indivisal, brands	Decisions makers Unit / Buying Center.
Airbus - Toulouse France	Buying 4 usage Center.
Air India - India, Etihad 4 Emirates → UAE.	geographically concentrated.

? Volume ??

Tilting Impact - lower | Tilting impact - higher.

Some basics.

- Human beings engage in exchange
- Exchange creates value.
- Value is created in a tangible manner (B2C/B2B

Amazon —sell product→ Ashwin.
Ashwin —receive currency→ Amazon

Value can also be created in an intangible manner! This is where service comes in!

Some basics

- Services - provision of economic value intangible in nature! Involves a form of rental through which customers obtain benefits.
- Examples: Education, transport - etc.
- Do we own it? Benefit of airline - transport moving us from one destination to another and provide in flight services.
- Rental services.

Why study services?

- Services contribute to almost 80% of the economy in developed countries, thereby dominating global economy.

- Most new jobs are generated by services.
- Provides firms with a competitive advantage - The SD/ servitization Eg. Rolls Royce. OSRs etc.

- Lets first understand the diff. between product & services.
  - Products are tigible / intangible.
  - In which case, is the pricing difficult?
  - Products are first produced, then sold and later consumed.
  - How about services then? Class participation.

Definition of Services (Wirtz and Lovelock, 2022):
Services are economic activities performed by one party to another. Often time based, these performances bring about desired results to recipients, object and assets.

In exchange for money, time and effort, service customers expect value from access to labour, skills expertise, good facilities, networks and systems. However they do not normally take ownership of the physical elements involved.

How product characteristics affect ease of evaluation



- High perceived risk
- Make consumers uncomfortable
- Doubts on service quality pricing etc.

- What can be done to drive away these?

What is STP? Segmentation Targeting & positioning.

Credence aspects → how it will help:.

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Organization Buying Behaviours (B2B purchase).

8 step process.

1. Problem recognition.
2. General description of the need/needs
3. Product specs
4. Supplier search
5. Analysis and acquisition of proposals.
6. Supplier selection
7. Selection of order routine.
8. Post usage performance reviews. → (common B2B, B2C.

B2B → less customers, purchase value is high, lot of people involved in decision making unit (DMU), purchase duration (3 months - 1 year). jilting effect is high.

1. Problem Recognition → In B2B, there is no impulse buying, In B2B, customers firms buy under 2 key conditions. (A) when there is problem, (B) when there is a need.

e.g. Engineers (Users) in a company complained that their laptops were slow and had a poor battery life.

2. General description of needs — How the company purchase dept/users, they list what is that they need. The engineers asked for faster high performing laptops and no having a good battery life.

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③ Product specs: Based on the need(s) identified in step 2, the specs of the product is provided. Laptops needed 250.

Processor (intel 7 or above), screen size (13 inches).
RFI (Request for Information).

④ Supplier search --- purchase dept based on market research identify potential suppliers. lets say identified 15 vendors.

» RfQ (Request for Quote)

⑤ Analysis and acquisition of proposals - here the members of the DMU analyse the quotes/proposals given by each supplier in detail.

⑥ Supplier Selections.

No of laptops = 250

hp	65000
dell	72K
lenevo	69.5 K
apple	80K

Price is not only the factor that drives purchase in B2B.

Purchase decisions rather are based on CSF/KSF.

Critical / Key success factors

CSF	Weightage	Rating (1-5)	
Price	10 %	5	0.5
Brandinage	20 %	4	0.8
Service quality	30 %	3	0.9
Flexibility	10 %	5	0.5
Order of quality	30 %	5	1.5
			4.2

Can be multiple factors.

Supplier selected : Dell.

① Selection of order Routine Agreement and contracts are always higher in B2B (Product value is high)

50 % cash & 50 % installments.

② Post usage performance reviews.

There are 3 diff buying situation in B2B.

① Straight rebuy
② Modified rebuy
③ New task situation.

Straight Rebuy

Done for products that inexpensive and/or bought frequently.

Don't have to do all steps.

New task Situation.

Done for products that are "expensive" and bought for the first time"

will have to do 8 steps.

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Modified Rebuy.
Some modification may be needed in the existing product.

» Depends on the cost.

Decision Making Unit (DMU) | Buying Centre | BUC - Buying & Usage centre.

B2B - the end customer is a "Bussiness"
Purchase duration is "longer" (3 months - 1 year)
Purchase value is higher (millions to billions ₹, $
Decision making body - DMU / BUC (5-15 people).

People within the DMU play diff roles
① Gatekeeper ② Users ③ Initiator
④ Finance, HR, Quality, Ops
⑤ Influencers ⑥ Decision maker.

Negative inheritence of domino effects.
Supplier → Customers → Clients

① Gatekeeper : security, Finance, HR, Quality, Ops.
② Users : are those people who actually uses the product/services and are the ones who identify the problem/ specify need (s).

③ Initiators : when users identify a problem and on have issued, they pass this information to the initiators. Supervisors, managers, purchase dept. etc.
④ Influencers : Head of dept, Director, VP etc. who are closer to decision maker.

⑤ Decision Maker : Head, CEO, Chief Purchase officer....

Steps of B2B professional selling.

Sales Techniques

Techniques / methods used by the salespeople to convince / persuade / woo the customers to buy.

Types of Sales Techniques

① Direct Selling (B2B) / (B2C)
Seller ...→ Buyer.

② Indirect Selling.
Seller → Intermidiate (Agent) → Buyer.

③ Whole Selling (B2B)
Manufacturer (Producers) → distributors → wholesellers → Retailers → Customers.

Manufacturer volume > distributor volume > wholeseller > retailers volume > Customer volume

Retail selling → Spencers etc
Online selling
Door to door selling (Obsolete)

Marketplace vs Marketspace.
↓
physical place where buyer & seller meet, Negotiations
virtual place, buyer & seller are connected by means of technology....

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Graded Assignment:

(13) AB Company

		Score	
Pricing	50%	5	2.5
Service quality	30%	4	1.2
Brand Image	10%	1	0.1
Market Reputation	10%	1	0.1
			3.9 Ans.

(2) When the products are purchased frequently despite the high prices refer to as : Straight Rebuy (Acc. to me.

wrong (1) When the organization make decisions on purchases, they tend to decide on.

Ashwin Price & quality is not only factor. There are other factors too. The correct answer is (SF/RSF, since you have to choose. none of the above.

(5) The Phenomenon of jilting is common in Ashwin B2B only

(12) Once the proposals were received, the internal department started analyzing the same to make a recommendation This is referred as.
Both Buying & selling Centre.

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Week-4 Consumer behaviour

Utility: when you're consuming goods & services so what is the satisfaction of benefit you are getting it from there, is called utility.

Concept of utility → (helps us to understand) → Consumer demand theory

How people make choice about what to buy based on the satisfaction they get from buying that goods / service.

Economist

How and why customers make choices??

Economists → Total utility (TU) → foundation of Consumer economics
→ marginal utility.

Total utility: overall amt of satisfaction which ya gets after consuming specific quantity of goods or services.

ex Total happiness is Consuming 3 slice of Pizza

↳ learning module 0 of BDM theory course

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Marginal Utility → additional satisfaction by consuming 1 more unit of goods and service.

Laws of Diminishing Marginal utility [LDMU]

Consumption ↑↑↑ Utility ↓↓↓

Assumptions

① No. of units are standard.
② Identical in nature.
③ Consumption process is without any gap
④ Consumer taste/habits; tems during consumption.
⑤ Income level should be constant
⑥ Price of Commodity should be constant

Aviation Price Management :

DMU. → Customer Psychology and customer lifetime value

case ① Flight delay (economic)
↳ 2 hrs late.
↳ food items to eat.
↳ satisfaction → Positive
↳ steps → satisfied.

MU - high

case ② Total delay → 4 hrs.
↳ coffe 4 snacks.
↳ MU is moderate & low

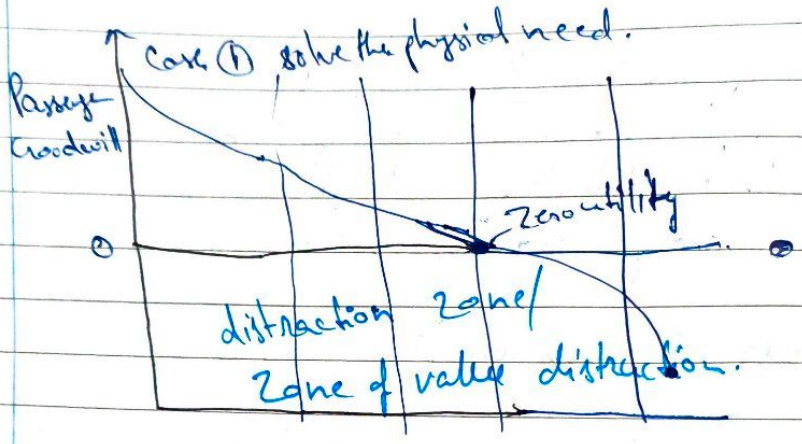


Case ① 3hrs know Total delay - 7hrs.
critical

↳ MU = 0

↳ Tutility = Dropen
⇓
mood.

Case ②: Total delay - 10hrs.
MU = -ve
Total utility - Hostile / angry



Delay 3>4 → failed path (CLV) destroyed
→ optimal path (butta deal) presence (CLV)

Pivot →

Mathematical Relationship:

$MU = \frac{\text{Change in Total utility}}{\text{change in quantity}}$

$MU = \frac{\Delta TU}{\Delta Q} = \frac{T_{n+1} - T_n}{\Delta Q}$

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Goods consumed → Airline compensation & Apologies
Utility - Passenger/ Customer goodwill (satisfaction)

Unit Compensation	Situation	Marginal utility (Passenger Goodwill)	Total Utility (Overall mkt.)
① Initial meal (Food) voucher	2 hrs delay	High	Stabilised/Positive
② Coffee + Snacks	4 hrs delay	moderate/low	flat
③ future 1K discount coupon	7 hrs.	Zero. You doesn't care abt future.	Dropping.
④	10 hrs.	Negative	

Supply & demand.

Water - diamond paradox. (Paradox of value.

suggests
→ have birth → To modern pricing theory.
→ Price → driven by marginal utility and scarcity.
⇓
Not by total utility.
⇓
Not by total utility.

Classical Economic Puzzle
⇓ highlights
disconnect b/w the use value and exchange value of goods

Water → cheaper / essential

Diamond → Non-essential → Too expensive.

Water :-
High Total Utility & Low MU

Total Utility
Need for survival ;
Utility is immense.

Marginal Utility ↓ Water is abundant
↳ easily available

Diamond

Economics → Price determination ∝ Not on Total Supply.
based on Marginal Utility

Diamond

→ Total utility = low → Luxury ornament.
Marginal Utility = high.

→ little functional use.
→ Rare.
↳ Supply Restriction
↳ Marginal utility becomes high.
Price Tag

Supply - Demand Interaction

Paradox :- How interaction of supply and demand determines price.

Goods	Supply	Marginal Demand	Price
Water	v. high	low (due to easily available)	low
Diamond	V. low	High due to scarcity	high

Key takeaways:

① Price Reflect = Marginal not Total

Market Price of Goods $\xrightarrow{\text{not depend on}}$ Humanitary Importance

Market Price of Goods $\xrightarrow{\text{depend on}}$ Satisfaction & usefulness of very next available (marginal utility).

Next unit of glass of water worth nothing
Next Diamond unit = work as a fortune.

① Value is subjective and context dependent.
② Scarcity → Engine of value.
Even life saving goods commands zero value (like air).

Paradox Prove: Human being only assign value/monetary value to things that are required to tradeoffs

Demand: Relationship b/w Price and Quantity demanded.

↳ Desire to buy → Purchasing power.

ability & willingness to buy

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Determinants of Demand.

- Price of Commodity
- Price related to substitutes. → Peas (A, B, C)
- Income of consumer.
- Taste and preference
- Wealth of consumer.

Derived function ⇒ $D_{x\,2} = f(P_A, P_B, P_x, T_2, I_2)$.

Law of demanded:

Negative / invers relationship b/w price & quantity.

(ex) Buy more ⇒ price ↓ | other things are
Buy less ⇒ price ↑ | constant.

Assumptions

- No change in income
- No change in consumer taste.
- Price of the goods No change.
- No substitute for the goods.

Demand Schedule & Demand Curve.

Demand schedule ⇒ Tabular statement.

Movement of demand Curve:

Moving in the same curve.

Case Study

5G Network & Hardware

Hardware → smart phone

Driven by consumers

↓

Ability to pay and willingness @ diff. price range

Supply side:

↳ M/g. Drives supply for consumers demand.

Services - 5G phone service

Demand side

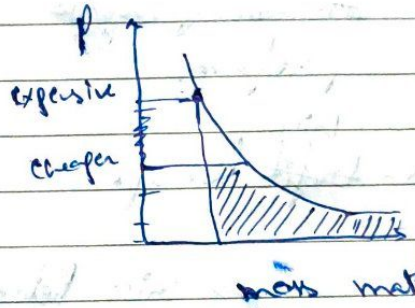
↳ high speed

↳ low cost

Service side

↳ Telecom operator

↳ Dictated by infrast development and spectrum availability.

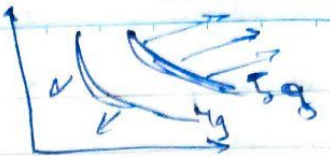


	Movement	Shift.
	Change in Goods price	No price variable changing.
		
5G smart phone	Retail price drops down due to scaling production	General consumer income rise / fall.
5G Data	Telecom Provider slashes monthly tariffs; increases data consumption	WFH ⇒ High speed data. (work from home)

Economic forces

① Network Plan
- Cost of usage
- Cost of data.

② Smartphone price



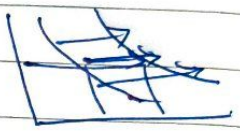
5G Market Equilibrium ⟶ Mobile healthcare



→ Smart Logistics



→ Streaming media platform



Demand Shift ⟶ Price

↳ Non-~~profit~~ Concept.

Movement ~~Shift~~ ⟶ Price Related Concept.

Along Demand Curve

⊛ Consumer-Income-change.

Demand of some goods: ⇒ Very responsive to change in Price.

⇒ Not Responsive to change in Price

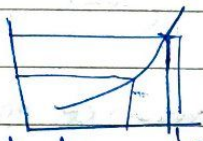
Price Elasticity of Demand: (PED)

What effects the elasticity of Demand ????

Consumers are price Sensitive.

	Change in Price	Change in Qd	ΔP%	ΔQ%	Revenue = P x Q	Ed
①	↑	↓	+10	-8	↑	Price inelastic
②	↑	↓	+10	-12	↓	Price Elastic
③	↑	↓	+10	-10	No change	Unit elastic
④	↓	↑	-10	+15	↑	Price Elastic
⑤	↓	↑	-10	+7	↓	Price Inelastic
⑥	↓	↑	-10	+10%	No change	Unit Elastic

Supply: S(Q, P)

Law of Supply  positive relationship

more production ——leads——→ more expensive to produce

⇓

because we have limited resources to produce goods/service

Need more compensation / money if unit price ↑

Opportunity Cost supporting positive Relationship.

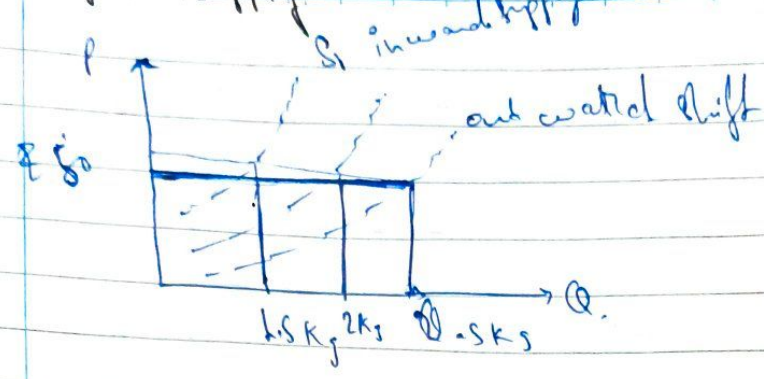
price 3k/hr — 10hr

price 5k/hr — 30hr

This implies ⇒ Positive Relationship b/w price and your supply

↳ price (wage) ↑ ⇒

Shift N Supply



Market Equillibrium

Price & Quantity demanded are equal.

QDemanded = Quantity Supply.

Law: Rate of Gold.

Suppose Rate of gold = £ 15K / gram.

Rate of gold driven by market equillibrium.

Amount of gold buyer wants to purchase

⇓

exactly meets to the amount price & seller utility to sell.

Two Economic forces ⇒ → Demand / 2 forces.
→ Supply

⇓

drive Prices of gold.

Demand Curve (buyers)

Supply Curve (sellers)

↳ miners, importer, Recycler

Law of Supply Price ↑↑ Of demanded ↑↑

Equilibrium meet kat.

Seller = ₹ 16K / gram (want to sell) → backoff ⇒ surplus (Excess inventory)

Seller = ₹ 13K / gram. → lower the price to make seller

↓ Shortage of gold.

↓ instantly he'll get a buyer.

Demand = $P = a - bQ$ → negative relation; slope

Supply = $P = C + dQ$ → Positive Relationship

Case Study of IPL

CSK final in Chennai

Management ⇒ determine / optimise the price of ticket.

Supply ⇒ fixed 50000 seats.

↳ Inelastic.

↑↑↑

Demand: The willingness to pay ⇒ Customs to Customs.

P ↑↑↑ Q_d ↓↓

P ↓↓ Q_d ↑↑

$Q_D = 1500 - 125P$

↳ Logic comes from data Analytics.

↳ Historical data.

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Market Equilibrium

$Q_s = Q_d$

$6000 = 15000 - 1.25P$

$P = 8000$ → premium tickets

↳ 60000 – full.

If price wrong

Management = 4K ↓ ⇒ P_{new}.

$Q_D = Q_S$

$15000 - 1.25P_{new} = Supply$

$S = 10,000$ → shortage of seats / supply

8000 shortage.

» Price too high

$P_{new} = 10,000$

$S = 2500$ tickets

↳ 2500 → (surplus; Tickets)

↓↓ empty seats.

④ Price Elasticity of Demand

price 8000 → 9600 (20% change)

$P_1 = 8000$

$P_2 = 9600$

$Q_2 = 15000 - 1.25(9600)$

$Q = 3000$

$\Delta P\% = 20\%$

$\Delta Q(\%) = \frac{3000 - 5000}{5000} = -40\%$

PED = 2 Elastic demand (> 1)

Perfectly Inelastic Segment:

$Q_D = 6000 - 0.12P$

$P_1 = 5000$ $Q_D = 5000$

$P_{new} = 9600$; $(Q_D)_{new} = 6000 - 0.12P$
$= 4800$ tickets

PED = 0.2
↳ Inelastic in nature.

Revenue 1

TR = P × D = 5000 × 8000 = 4 cr.
TR @ hike hike = 4800 × 6000 = 4.6 cr.

Demand = Inelastic ∴ hike increases Total Revenue.

$Q_D = 13000 - 1.0P$
$P_{old} = 8000$
$P_{new} = 8200$
$(Q_D)_{new} = 4200$
% change $Q_2 = -16\%$

PED = 1.6% > 1 Demand are elastic

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TR = 4cr.
TR = 8800 × 4200
= 3.6 cr.

Demand elastic - Price ↑ then Total Revenue ↓

	Economic Concept	IPL applied	Numerical Example
①	Demand	willing to buy tickets and waiting	$Q_D = 15000 - 1.25P$
②	Supply	fixed Capacity of Stadium	$S = 5000$ fixed
③	Market Equilibrium	No black money; All seats are booked ⇓ Perfect point / equilibrium point	$\$ 8000$ } House $S_2 = 5000$ }
④	Inelastic	VIP / cricket diehard fan	$PED = 0.2$ $P \uparrow TR \uparrow$
	Elastic	TV	$PED = 1.6$ $P \uparrow TR \downarrow$

Q "Elastic in the short run"

The no. of seats are fixed!!